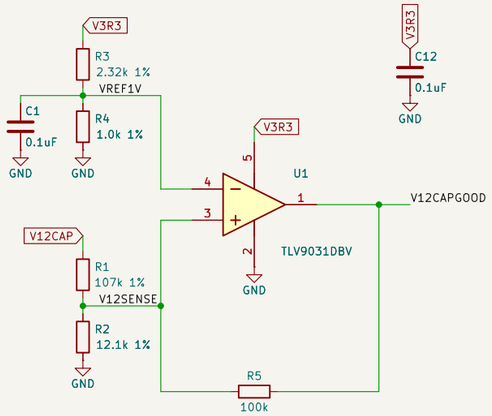
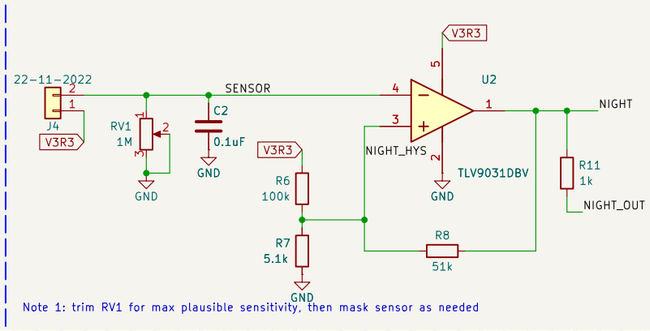
## V12CAP Monitor (U1)

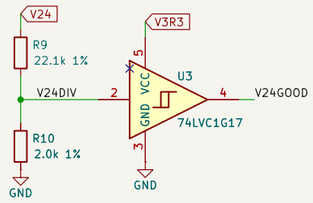



## Night Sensor (U2, J4)

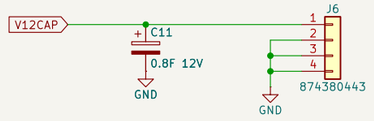



Note 1: trim RV1 for max plausible sensitivity, then mask sensor as needed

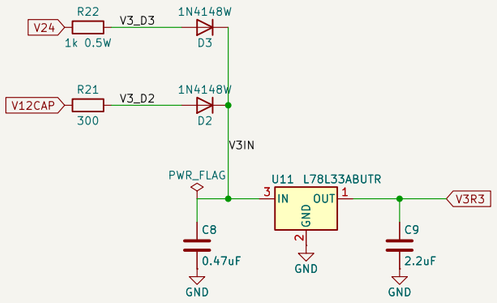
## V24 Detect (U3)



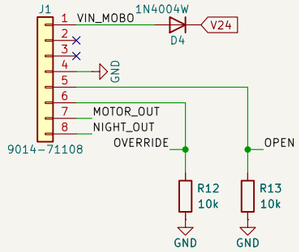

## Backup Capacitor (C11, J6)



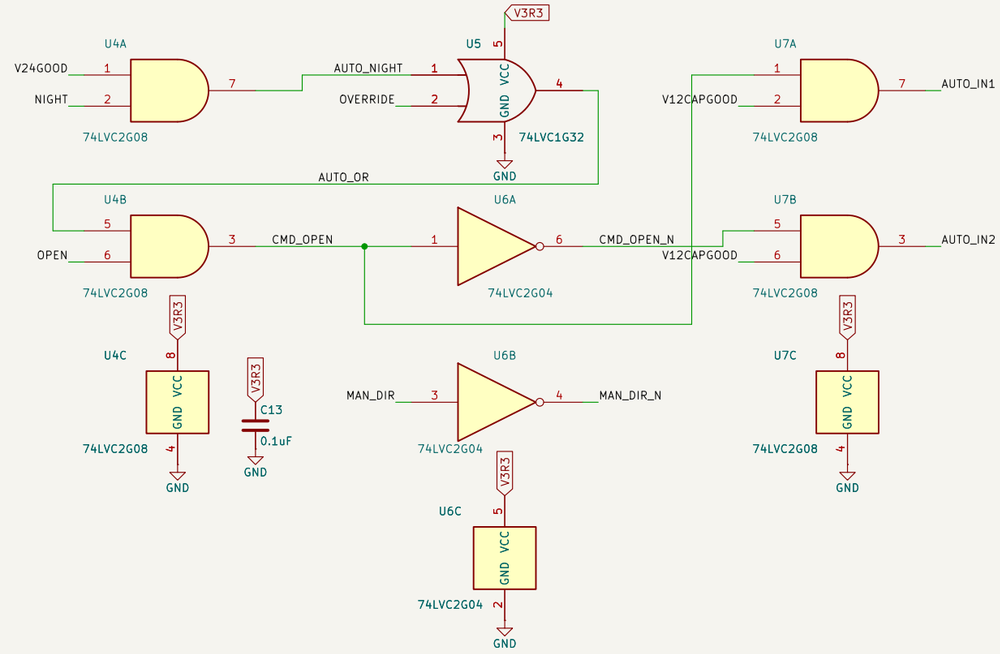

## 3V3 Regulator (U11)



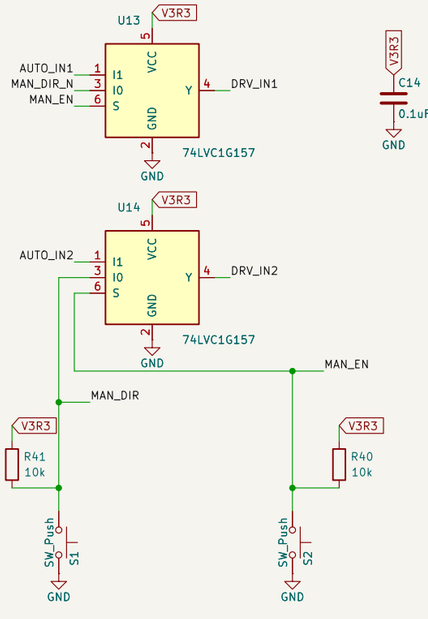

## MOBO Connector (J1)



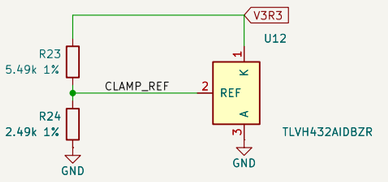

## Control Logic (U4–U7)



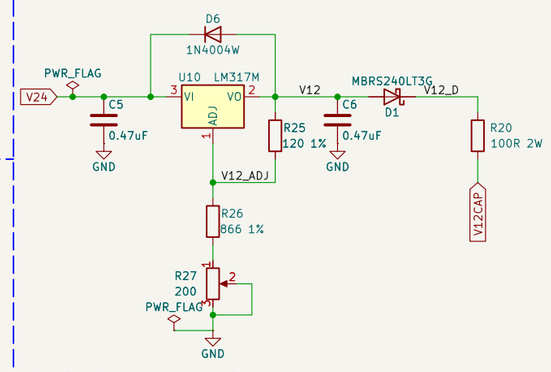

## Manual/Auto Select (U13, U14)




## 3V3 Clamp (U12)

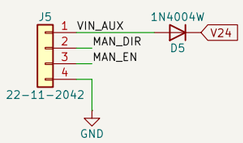



## 12V Regulator (U10)

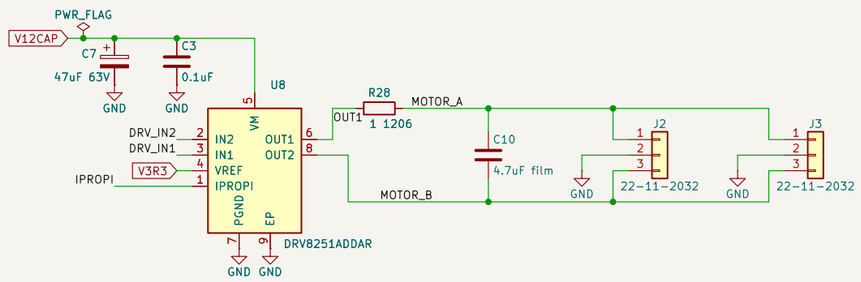



Note 3: U10 heatsink area for 1.5W peak:
suggest min 1 cm^2, prefer 12mm x 12mm or equiv.
pin 2 may be absent

## Aux Power / Manual (J5)



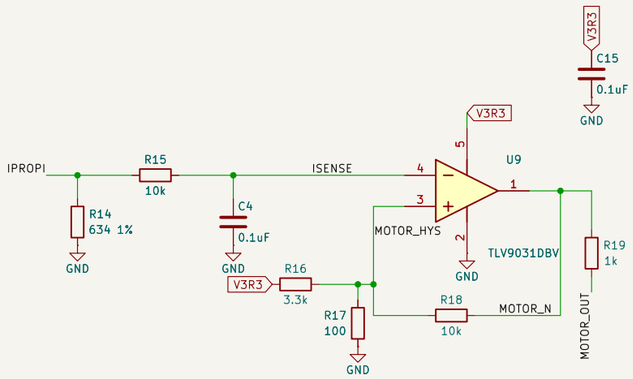

## Motor Driver (U8, J2, J3)




**DRV8251A truth table**

| IN1 | IN2 | OUT1 | OUT2 |
|---|---|---|---|
| 0 | 0 | Hi-Z | Hi-Z |
| 0 | 1 | L | H |
| 1 | 0 | H | L |
| 1 | 1 | L | L |

## Current Sense (U9)




Note 2: IPROPI Iout = 1.58 mA/A; node internally clamped to V<=Vref;
motor current limits when V=Vref, here Ilim=3.3A

Sheet: /
File: shuttercontrol.kicad_sch

**Title: New Shutter Controller**

| Size: A3 | Date: 2026-11-02 | **Rev: v.02j** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 1/1 |